



CS & IT ENGINEERING



COMPILER DESIGN

PARSING

Lecture No.- 05



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Recap of Previous Lecture



Topic

✓ First set
??

Follow set

SLR(1) Parser



Topics to be Covered



Topic

??

Follow set

Topic

??

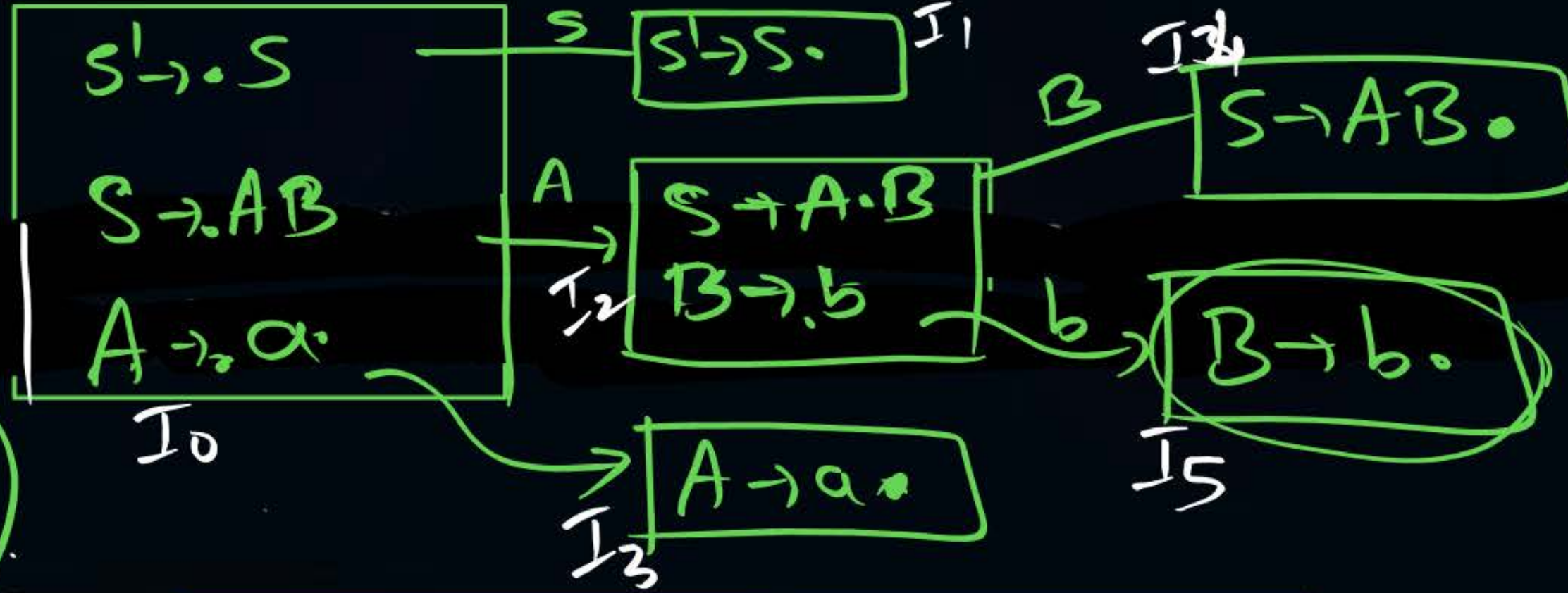
Topic

??

LR(0) item DFA



- ① $S \rightarrow AB$
- ② $A \rightarrow a$
- ③ $B \rightarrow b$



LR(0) & SLR(1)

SLR(1) Parsing Table

	a	b	\$	S	A	B
I ₀	S ₃			1	2	
I ₁			Accept			
I ₂		S ₅				4
I ₃		r ₂				
I ₄			r ₁			
I ₅			r ₃			

Follow(A) = {b}

$A \rightarrow a \cdot$ r₂

Follow(S) = {\$}

$B \rightarrow b \cdot$ r₃
Follow(B) = {\$}

① Shift action $\Rightarrow$ Same

② Goto entries $\Rightarrow$ Same

③ Reduce actions may be less in SLR(1) Table

SLR(1) Grammar Detection

Reduce/Reduce Conflict

①

$A \rightarrow \alpha \cdot$
 $B \rightarrow \gamma \cdot$

② $\text{Follow}(A) \cap \text{Follow}(B) \neq \emptyset$

Shift/Reduce Conflict

①

$A \rightarrow \alpha \cdot a B$ (Shift)
 $B \rightarrow \gamma \cdot$ (Reduce)

terminal

② If $a \in \text{Follow}(B)$ then S/R Conflict



Topic : First Set

1. $\text{First}\{abc\}=\{a\}$

2. $A \rightarrow aX$

$$\text{First}(A)=\{a\}$$

3. $A \rightarrow Bx$

$$\text{First}(A)=\text{First}(B)$$

4. $A \rightarrow BX$

$$\text{First}(A)=\{\text{First}(B)-\epsilon\} \cup \{\text{First}(X)\}$$



Topic : Follow Set

of a Non terminal is set of terminals which appears immediately R.H.S of that N.T in some Sentential form.

1. ✓

$$\text{Follow}(s) = \{\$ \}$$

2. ✓

If Grammar is

$$A \rightarrow \alpha B X$$

$$X \rightarrow \epsilon$$

$$\text{Follow}(B) = \text{First}(X)$$

$$\begin{array}{c} A _ \\ \uparrow \\ \alpha B _ \end{array}$$

3. ✓

If Grammar is

$$A \rightarrow \alpha B _$$

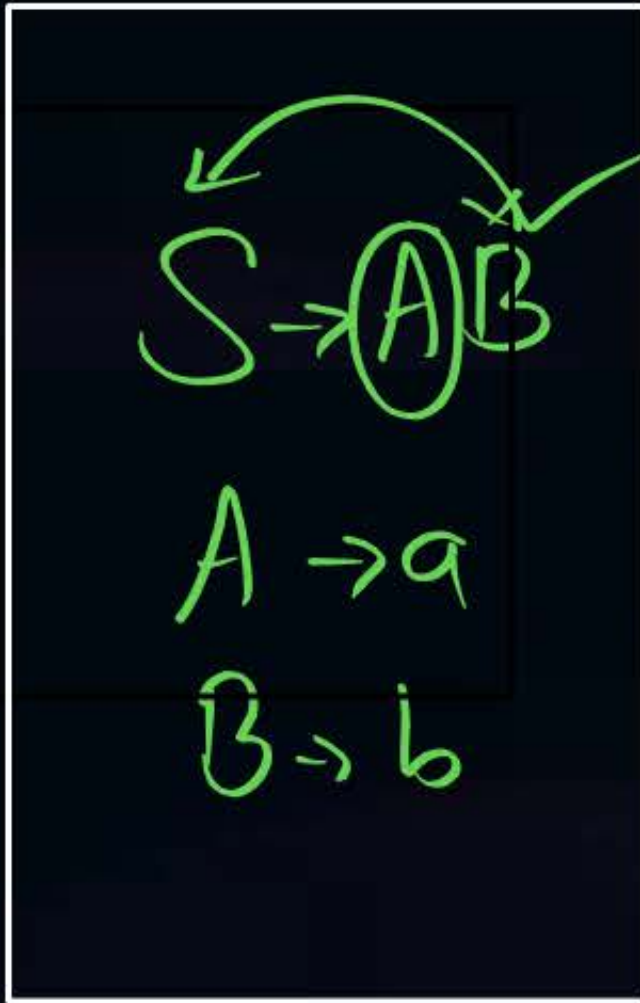
$$\text{Follow}(B) = \text{Follow}(A)$$

4. ✓

If Grammar is

$$A \rightarrow B X \quad X \rightarrow \epsilon$$

$$\text{Follow}(B) = \{\text{First}(X) - \epsilon\} \cup \text{Follow}(A)$$



L.H.D

1) $(S) \$$ ✓

2) $A (B) \$$

3) $a B \$$ ✓

4) $\underline{a} \underline{b} \$$

R.M.D

5) $(S) \$$ ✓

6) $A \underline{B} \$$

7) $(A) (b) \$$

8) $a b \$$

✓ Follow(S) = { \$ }

✓ Follow(B) = { \$ }

✓ Follow(A) = { b } ✓

$$S \rightarrow \underline{A} \underline{a} \underline{A} \underline{b} \mid \underline{B} \underline{b} \underline{B} \underline{a}$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$\text{Follow}(S) = \{ \$ \}$$

$$\text{Follow}(A) = \{ a, b \}$$

$$\text{Follow}(B) = \{ b, a \}$$

$$S \rightarrow A \underline{a} \mid b A \underline{c} \mid d$$

$$A \rightarrow d$$

$$\text{Follow}(S) = \{\$ \}$$

$$\text{Follow}(A) = \{a, c\}$$

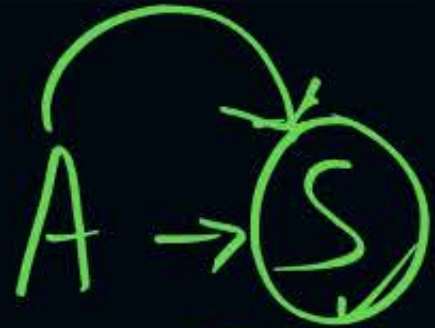
$$\textcircled{3} \quad S \rightarrow a \underline{S} \textcircled{A} | \epsilon$$

$$A \rightarrow c | \epsilon$$

$$\text{Follow}(S) = \underline{\underline{\{\$, c\}}}$$

$$\text{Follow}(A) = \underline{\underline{\{\$, c\}}}$$

④ $S \rightarrow aA \underline{bB} \mid bA \underline{aB} \mid \epsilon$



$$\text{Follow}(A) = \{a, b\}$$

$$\text{Follow}(S) = \{\$, a, b\}$$

$$\text{Follow}(B) = \{\$, a, b\}$$

⑤ $S \rightarrow L = \overset{\curvearrowright}{\textcircled{R}}$

$S \rightarrow \overset{\curvearrowright}{\textcircled{R}}$

$L \rightarrow * \overset{\curvearrowright}{\textcircled{R}}$

$L \rightarrow a$

$R \rightarrow \overset{\curvearrowright}{\textcircled{L}}$

$$\text{Follow}(S) = \{ \$ \}$$

$$\text{Follow}(L) = \{ =, \$ \}$$

$$\text{Follow}(R) = \{ \$, = \}$$

$$\text{FOLL}(T) = \text{FIRST}(E') - \{+, \epsilon\}$$

⑥ $E \rightarrow \underline{T} E'$

$$E' \rightarrow + \underline{T} E' \mid \epsilon$$

$$T \rightarrow \underline{F} T'$$

$$T \rightarrow * \underline{F} T' \mid \epsilon$$

$$F \rightarrow (\underline{E}) \mid a$$

$$\text{FOLL}(F) = \text{FIRST}(T') = \{*, \epsilon\}$$

	<u>Follow</u>
$\text{FOLL}(E)$	$\{\$, \epsilon\}$
$\text{FOLL}(E')$	$\{\$, \epsilon\}$
$\text{FOLL}(T)$	$\{+, \$, \epsilon\}$
$\text{FOLL}(T')$	$\{+, \$, \epsilon\}$
$\text{FOLL}(F)$	$\{*, +, \$, \epsilon\}$

THANK - YOU